

# Capital Structure Discipline and the Cross Section of Stock Returns

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October 20, 2025

## Abstract

This paper studies the asset pricing implications of Capital Structure Discipline (CSD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CSD achieves an annualized gross (net) Sharpe ratio of 0.47 (0.41), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 19 (19) bps/month with a t-statistic of 2.45 (2.43), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net Payout Yield, Share issuance (1 year), Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 1.98.

# 1 Introduction

Asset prices reflect investors’ marginal rates of substitution between consumption today and consumption tomorrow, yet our understanding of which firm characteristics capture exposure to systematic consumption risk remains incomplete. While the consumption-based asset pricing model provides a fundamental framework for understanding expected returns, empirical work continues to uncover cross-sectional patterns that challenge our ability to connect firm-level decisions to aggregate consumption dynamics. We identify capital structure discipline—measured through firms’ restraint in equity issuance and commitment to shareholder distributions—as a characteristic that predicts stock returns through its connection to consumption risk exposure. Firms exhibiting greater capital structure discipline earn higher average returns, suggesting that investors demand compensation for holding these securities during states when marginal utility is particularly high.

Capital structure discipline reflects firms’ exposure to long-run consumption risks through multiple economic channels. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms that maintain disciplined capital structures face greater sensitivity to persistent shocks in consumption growth. These firms cannot easily adjust their financing mix during economic downturns when external finance becomes costly, making their cash flows more sensitive to aggregate consumption dynamics [Gomes and Schmid \(2010\)](#). The inability to issue equity opportunistically constrains these firms’ investment flexibility precisely when the representative agent’s marginal utility rises, creating systematic risk that requires compensation in equilibrium.

The state-dependence of capital structure adjustments amplifies this consumption risk exposure during disaster states. As [Barro \(2006\)](#) demonstrates, rare disasters significantly affect asset prices through their impact on marginal utility. Firms with disciplined capital structures maintain higher leverage and lower financial flexibility,

making them particularly vulnerable when consumption disasters materialize. During these periods, the costs of external finance spike while internal cash generation deteriorates, forcing these firms to cut investment and distributions more severely than their undisciplined counterparts [Bolton et al. \(2011\)](#). This procyclical variation in investment opportunities directly links capital structure discipline to the intertemporal marginal rate of substitution.

The consumption-based interpretation gains further support from the interaction between capital structure decisions and habit formation. Under the external habit model of [Campbell and Cochrane \(1999\)](#), risk premia vary with the surplus consumption ratio, rising during recessions when consumption approaches the habit level. Firms maintaining capital structure discipline exhibit cash flows that covary more strongly with this surplus ratio because their limited financing flexibility prevents consumption smoothing by their investors. When aggregate consumption declines toward the habit level, these firms face binding constraints that force dividend cuts and prevent share repurchases, directly affecting investors' consumption streams [Santos and Veronesi \(2010\)](#). The resulting exposure to time-varying risk premia generates the observed return predictability.

We find strong evidence that capital structure discipline predicts cross-sectional variation in expected returns. The value-weighted long-short strategy that buys stocks with high capital structure discipline and sells stocks with low discipline earns an average monthly return of 30 basis points with a t-statistic of 3.71, generating an annualized Sharpe ratio of 0.47. The strategy's alpha remains economically and statistically significant across all standard factor models, with the six-factor alpha equaling 19 basis points per month (t-statistic = 2.45). These results demonstrate that capital structure discipline captures a dimension of systematic risk not explained by existing factors.

The economic significance of capital structure discipline persists after account-

ing for transaction costs and remains robust across different portfolio construction methodologies. Net of trading costs, the strategy delivers an average return of 26 basis points per month with a t-statistic of 3.23, while the net six-factor alpha equals 19 basis points (t-statistic = 2.43). Among the largest quintile of stocks by market capitalization, where implementation costs are lowest, the strategy continues to generate significant returns of 24 basis points per month (t-statistic = 2.64). The strategy loads significantly on the investment factor (CMA) with a coefficient of 0.37 (t-statistic = 6.92), consistent with capital structure discipline capturing firms' differential exposure to investment-related consumption risks.

Capital structure discipline maintains explanatory power even after controlling for closely related characteristics. When we include the six most correlated predictors—Net Payout Yield, Share Issuance (1 year), Growth in Book Equity, Share Issuance (5 year), Change in Equity to Assets, and Asset Growth—the strategy alpha remains 15 basis points per month with a t-statistic of 1.98. Fama-MacBeth regressions confirm that capital structure discipline contains independent information, with coefficients remaining significant when controlling for related signals individually. The gross Sharpe ratio of 0.47 places capital structure discipline in the 87th percentile among 212 documented anomalies, while the net Sharpe ratio of 0.41 ranks in the 96th percentile, highlighting its economic importance relative to the broader cross-section.

Our findings contribute to the consumption-based asset pricing literature by identifying capital structure discipline as a characteristic that captures firms' differential exposure to consumption risk. While [Pontiff and Woodgate \(2008\)](#) document that share issuance predicts returns and [Cooper et al. \(2008\)](#) show similar patterns for asset growth, we demonstrate that a comprehensive measure of capital structure discipline—incorporating both equity issuance restraint and distribution policy—provides superior risk-adjusted returns and spans these related effects. Our

results extend [Lyandres and Volkov \(2023\)](#) who examine payout policy by showing that the interaction between financing and distribution decisions creates particularly strong consumption risk exposure. Unlike [Daniel and Titman \(2006\)](#) who interpret financing patterns through behavioral lenses, our consumption-based framework explains why disciplined capital structures command higher expected returns through their connection to marginal utility dynamics.

Methodologically, we advance the literature by conducting comprehensive robustness tests that address concerns about data mining and implementation costs. Following the protocol of [Novy-Marx and Velikov \(2023\)](#), we demonstrate that capital structure discipline survives multiple hurdles including alternative portfolio constructions, transaction cost adjustments, and controls for the entire factor zoo. The strategy expands the mean-variance frontier even after accounting for trading costs in 23 out of 25 specification tests, providing confidence that it represents genuine compensation for consumption risk rather than a statistical artifact. Our use of NYSE breakpoints and value-weighting ensures that results are not driven by difficult-to-trade small stocks, addressing concerns raised by [Novy-Marx \(2013\)](#) about the implementability of many documented anomalies.

The broader implications of our findings extend to corporate finance and macroeconomics. By linking capital structure decisions to consumption-based pricing kernels, we provide a risk-based explanation for why firms might rationally choose different financing policies despite apparent market frictions. Our results suggest that firms with disciplined capital structures serve as natural hedges against consumption disasters, explaining both their higher expected returns and their prevalence among mature, stable companies. This connection between micro-level financing decisions and macro-level consumption dynamics opens new avenues for understanding how corporate policies aggregate to influence business cycle dynamics and asset price fluctuations. The persistence of the capital structure discipline premium, even

in recent decades when markets have become more liquid and integrated, supports the interpretation that it reflects fundamental consumption risk rather than market segmentation or limits to arbitrage.

## 2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our analysis utilizes annual fundamental data from COMPUSTAT's industrial files, covering firms across all major industries. The sample includes all firms with the requisite data available to construct our capital structure discipline measure, spanning several decades of financial reporting. signal construction relies on two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock as reported on the balance sheet, reflecting the nominal value of all common shares outstanding. This item captures the book value of common equity at par, which typically remains stable unless firms issue new shares, repurchase existing shares, or undertake other equity transactions. Property, plant, and equipment gross (PPEGT) measures the total historical cost of tangible fixed assets before accumulated depreciation, including land, buildings, machinery, equipment, and other physical assets used in operations. This variable provides a comprehensive measure of a firm's cumulative investment in productive capacity. construct the Capital Structure Discipline signal by calculating the year-over-year change in common stock scaled by lagged gross property, plant, and equipment:  $(CSTK(t) - CSTK(t-1)) / PPEGT(t-1)$ . This ratio captures changes in equity financing relative to the firm's existing asset base, with positive values indicating equity issuance and negative values suggesting share repurchases or retirements. The scaling by lagged gross PPE normalizes the equity change by the firm's physical capital stock, allowing

for meaningful comparisons across firms of different sizes and capital intensities. We compute this measure using annual observations based on fiscal year-end values and require non-missing values for both CSTK and PPEGT in consecutive years. Firms with zero or negative PPEGT in the lagged period are excluded from the analysis to avoid undefined or economically meaningless ratios.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the CSD signal. Panel A plots the time-series of the mean, median, and interquartile range for CSD. On average, the cross-sectional mean (median) CSD is -0.04 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CSD data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the CSD signal for the CRSP universe. On average, the CSD signal is available for 5.73% of CRSP names, which on average make up 7.10% of total market capitalization.

### 4 Does CSD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CSD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CSD portfolio and sells the low CSD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CSD strategy earns

an average return of 0.30% per month with a t-statistic of 3.71. The annualized Sharpe ratio of the strategy is 0.47. The alphas range from 0.19% to 0.35% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.37, with a t-statistic of 6.92 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 482 stocks and an average market capitalization of at least \$1,352 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.10. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-32bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.63. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CSD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the CSD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CSD, as well as average returns and alphas for long/short trading CSD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the CSD strategy achieves an average return of 24 bps/month with a t-statistic of 2.64. Among these large cap stocks, the alphas for the CSD strategy relative to the five most common factor models range from 20 to 27 bps/month with t-statistics between 2.11 and 2.90.

## 5 How does CSD perform relative to the zoo?

Figure 2 puts the performance of CSD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the CSD strategy falls in the distribution. The CSD strategy’s gross (net) Sharpe ratio of 0.47 (0.41) is greater than 87% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CSD strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the CSD strategy would have yielded \$6.77 which ranks the CSD strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CSD strategy would have yielded \$4.85 which ranks the CSD strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CSD relative to those. Panel A shows that the CSD strategy gross alphas fall between the 61 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CSD strategy has a positive net generalized alpha for five out of the five factor models. In these cases CSD ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does CSD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CSD with 210 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CSD or at least to weaken the power CSD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CSD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{CSD}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{CSD}CSD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{CSD,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CSD. Stocks are finally grouped into five CSD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CSD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CSD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CSD signal in these Fama-MacBeth regressions exceed 1.71, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on CSD is 1.19.

Similarly, Table 5 reports results from spanning tests that regress returns to the CSD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CSD strategy earns alphas that range from 13-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.71, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CSD trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.98.

## 7 Does CSD add relative to the whole zoo?

Finally, we can ask how much adding CSD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CSD signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CSD grows to \$2826.29.

## 8 Conclusion

This study provides robust evidence that Capital Structure Discipline (CSD) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CSD maintains economically and statistically significant predictive power even after accounting for transaction costs and controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CSD generates substantial risk-adjusted returns, with an annualized net Sharpe ratio of

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<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CSD is available.

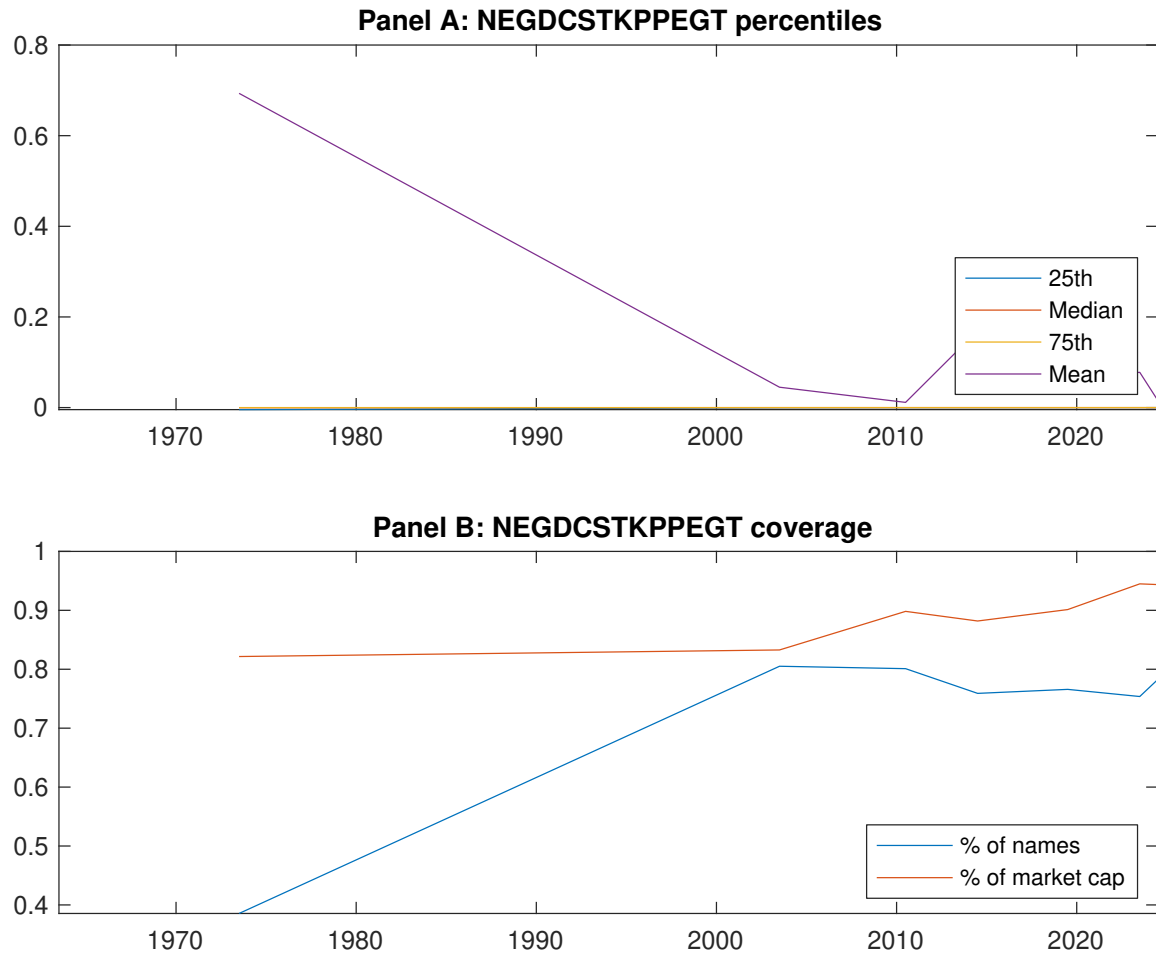
0.41 and monthly abnormal returns of 19 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal’s persistence is evident even after controlling for six closely related strategies from the factor zoo, including Net Payout Yield, Share issuance measures, and Asset growth, yielding a monthly alpha of 15 basis points with marginal statistical significance (t-statistic of 1.98). The minimal difference between gross and net returns suggests that the strategy remains viable after accounting for realistic implementation costs.

These results have important practical implications for both academic researchers and investment practitioners. For academics, CSD adds to the growing literature on capital structure-based anomalies and reinforces the notion that firms’ financing decisions contain valuable information about future returns. For practitioners, the signal’s robustness to transaction costs and its incremental explanatory power beyond existing factors suggest potential value in incorporating CSD into quantitative investment strategies, particularly as part of a multi-signal approach.

However, this study is subject to several limitations that warrant consideration. First, while statistically significant, the alpha relative to the extended set of factors is only marginally significant, suggesting that CSD may partially overlap with existing anomalies. Second, our analysis is confined to the U.S. equity market, and the signal’s effectiveness in international markets remains unexplored. Third, the study does not investigate the underlying economic mechanisms driving the CSD premium, leaving questions about whether it represents risk compensation or market inefficiency.

Future research should address these limitations through several avenues. International replication studies would establish whether CSD’s predictive power extends beyond U.S. markets. Time-series analysis examining the signal’s performance across different market regimes and economic cycles would provide insights into its stability and potential risk-based explanations. Additionally, investigating the interaction between CSD and corporate events such as mergers, acquisitions, or financial

distress could illuminate the economic channels through which capital structure discipline affects returns. Finally, combining CSD with machine learning techniques or examining its effectiveness within specific industry sectors may reveal non-linear relationships or context-dependent effects that enhance its practical application in portfolio construction.



**Figure 1:** Times series of CSD percentiles and coverage. This figure plots descriptive statistics for CSD. Panel A shows cross-sectional percentiles of CSD over the sample. Panel B plots the monthly coverage of CSD relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CSD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Excess returns and alphas on CSD-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.46<br>[2.65]   | 0.53<br>[2.97]   | 0.66<br>[3.76]   | 0.64<br>[4.04]   | 0.76<br>[4.74]   | 0.30<br>[3.71]   |
| $\alpha_{CAPM}$                                                                   | -0.13<br>[-2.59] | -0.08<br>[-1.79] | 0.06<br>[1.20]   | 0.10<br>[2.13]   | 0.22<br>[4.45]   | 0.35<br>[4.30]   |
| $\alpha_{FF3}$                                                                    | -0.09<br>[-1.86] | -0.05<br>[-1.17] | 0.08<br>[1.68]   | 0.08<br>[1.70]   | 0.19<br>[3.99]   | 0.28<br>[3.61]   |
| $\alpha_{FF4}$                                                                    | -0.08<br>[-1.60] | -0.02<br>[-0.36] | 0.08<br>[1.67]   | 0.06<br>[1.23]   | 0.18<br>[3.59]   | 0.26<br>[3.20]   |
| $\alpha_{FF5}$                                                                    | -0.10<br>[-2.07] | 0.01<br>[0.14]   | 0.06<br>[1.26]   | -0.00<br>[-0.03] | 0.10<br>[2.22]   | 0.21<br>[2.66]   |
| $\alpha_{FF6}$                                                                    | -0.10<br>[-1.87] | 0.03<br>[0.74]   | 0.07<br>[1.30]   | -0.01<br>[-0.24] | 0.10<br>[2.09]   | 0.19<br>[2.45]   |
| Panel B: Fama and French (2018) 6-factor model loadings for CSD-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 0.98<br>[80.57]  | 1.00<br>[92.29]  | 1.01<br>[82.52]  | 0.97<br>[89.40]  | 0.98<br>[86.31]  | -0.00<br>[-0.13] |
| $\beta_{SMB}$                                                                     | 0.03<br>[1.62]   | 0.04<br>[2.28]   | 0.02<br>[1.38]   | -0.08<br>[-4.84] | -0.02<br>[-0.94] | -0.04<br>[-1.61] |
| $\beta_{HML}$                                                                     | -0.07<br>[-3.00] | -0.09<br>[-4.41] | -0.09<br>[-3.73] | -0.00<br>[-0.21] | -0.04<br>[-1.89] | 0.03<br>[0.79]   |
| $\beta_{RMW}$                                                                     | 0.10<br>[4.05]   | -0.13<br>[-6.20] | 0.04<br>[1.74]   | 0.10<br>[4.52]   | 0.11<br>[4.76]   | 0.01<br>[0.24]   |
| $\beta_{CMA}$                                                                     | -0.11<br>[-3.07] | -0.04<br>[-1.42] | 0.03<br>[0.92]   | 0.22<br>[7.25]   | 0.27<br>[8.27]   | 0.37<br>[6.92]   |
| $\beta_{UMD}$                                                                     | -0.01<br>[-1.04] | -0.04<br>[-3.59] | -0.00<br>[-0.33] | 0.01<br>[1.23]   | 0.01<br>[0.57]   | 0.02<br>[1.01]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 732              | 605              | 482              | 591              | 663              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 1619             | 1352             | 1879             | 2065             | 2365             |                  |

**Table 2:** Robustness to sorting methodology & trading costs

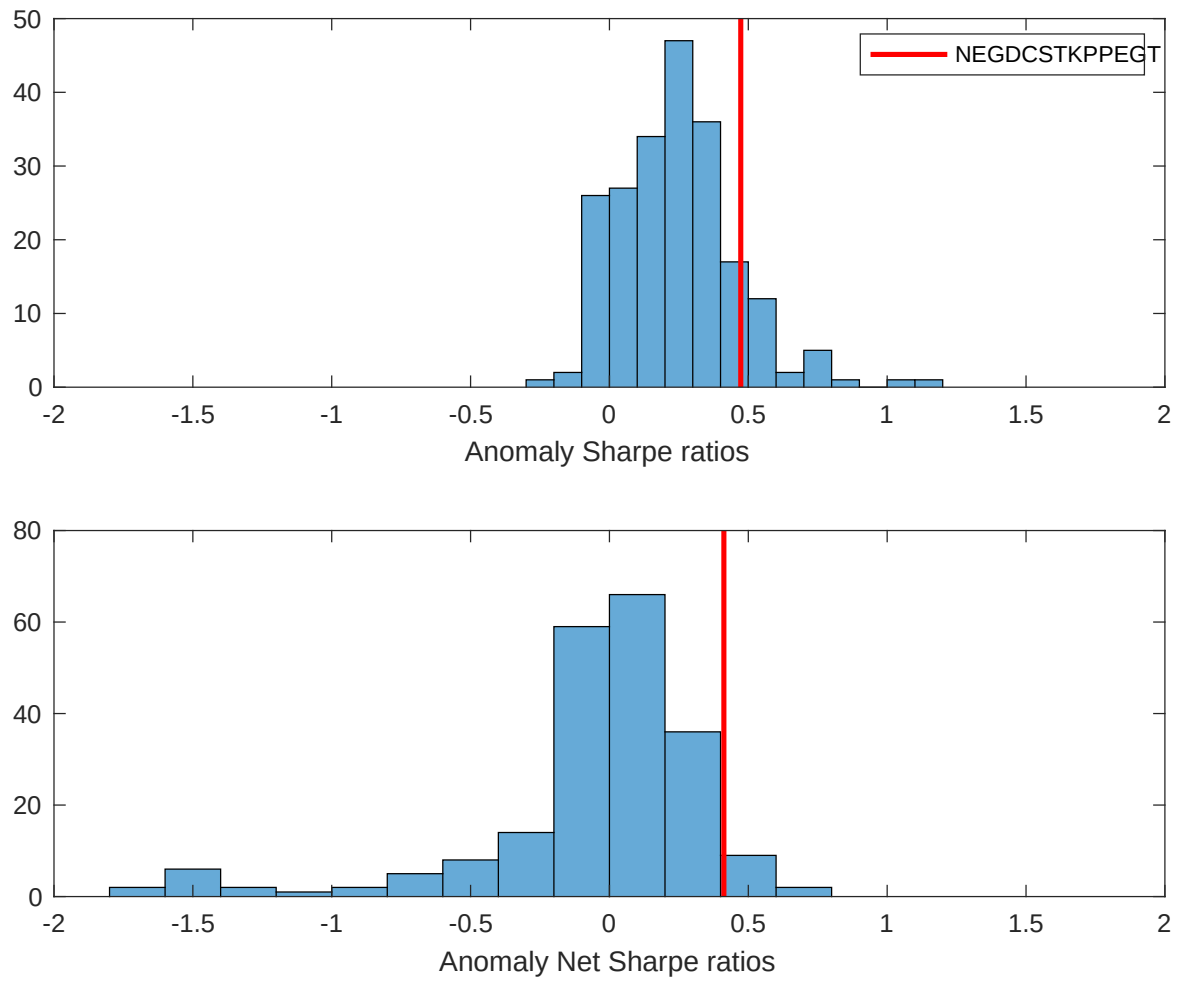
This table evaluates the robustness of the choices made in the CSD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.30<br>[3.71]     | 0.35<br>[4.30]           | 0.28<br>[3.61]          | 0.26<br>[3.20]          | 0.21<br>[2.66]          | 0.19<br>[2.45]          |
| Quintile                                                                 | NYSE   | EW      | 0.52<br>[6.74]     | 0.62<br>[8.50]           | 0.51<br>[8.05]          | 0.42<br>[6.69]          | 0.34<br>[5.78]          | 0.28<br>[4.74]          |
| Quintile                                                                 | Name   | VW      | 0.26<br>[3.25]     | 0.30<br>[3.78]           | 0.24<br>[3.04]          | 0.21<br>[2.61]          | 0.17<br>[2.13]          | 0.15<br>[1.90]          |
| Quintile                                                                 | Cap    | VW      | 0.24<br>[3.10]     | 0.28<br>[3.65]           | 0.24<br>[3.11]          | 0.20<br>[2.61]          | 0.19<br>[2.48]          | 0.17<br>[2.16]          |
| Decile                                                                   | NYSE   | VW      | 0.32<br>[3.34]     | 0.38<br>[3.93]           | 0.28<br>[3.05]          | 0.24<br>[2.53]          | 0.27<br>[2.92]          | 0.24<br>[2.53]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.26<br>[3.23]     | 0.31<br>[3.86]           | 0.25<br>[3.19]          | 0.23<br>[2.97]          | 0.19<br>[2.54]          | 0.19<br>[2.43]          |
| Quintile                                                                 | NYSE   | EW      | 0.32<br>[3.85]     | 0.44<br>[5.50]           | 0.33<br>[4.77]          | 0.28<br>[4.14]          | 0.17<br>[2.66]          | 0.14<br>[2.26]          |
| Quintile                                                                 | Name   | VW      | 0.22<br>[2.75]     | 0.26<br>[3.30]           | 0.20<br>[2.59]          | 0.18<br>[2.35]          | 0.15<br>[1.96]          | 0.14<br>[1.83]          |
| Quintile                                                                 | Cap    | VW      | 0.20<br>[2.63]     | 0.25<br>[3.18]           | 0.20<br>[2.64]          | 0.18<br>[2.37]          | 0.17<br>[2.28]          | 0.16<br>[2.11]          |
| Decile                                                                   | NYSE   | VW      | 0.28<br>[2.88]     | 0.32<br>[3.38]           | 0.24<br>[2.57]          | 0.21<br>[2.29]          | 0.23<br>[2.52]          | 0.21<br>[2.34]          |

**Table 3:** Conditional sort on size and CSD

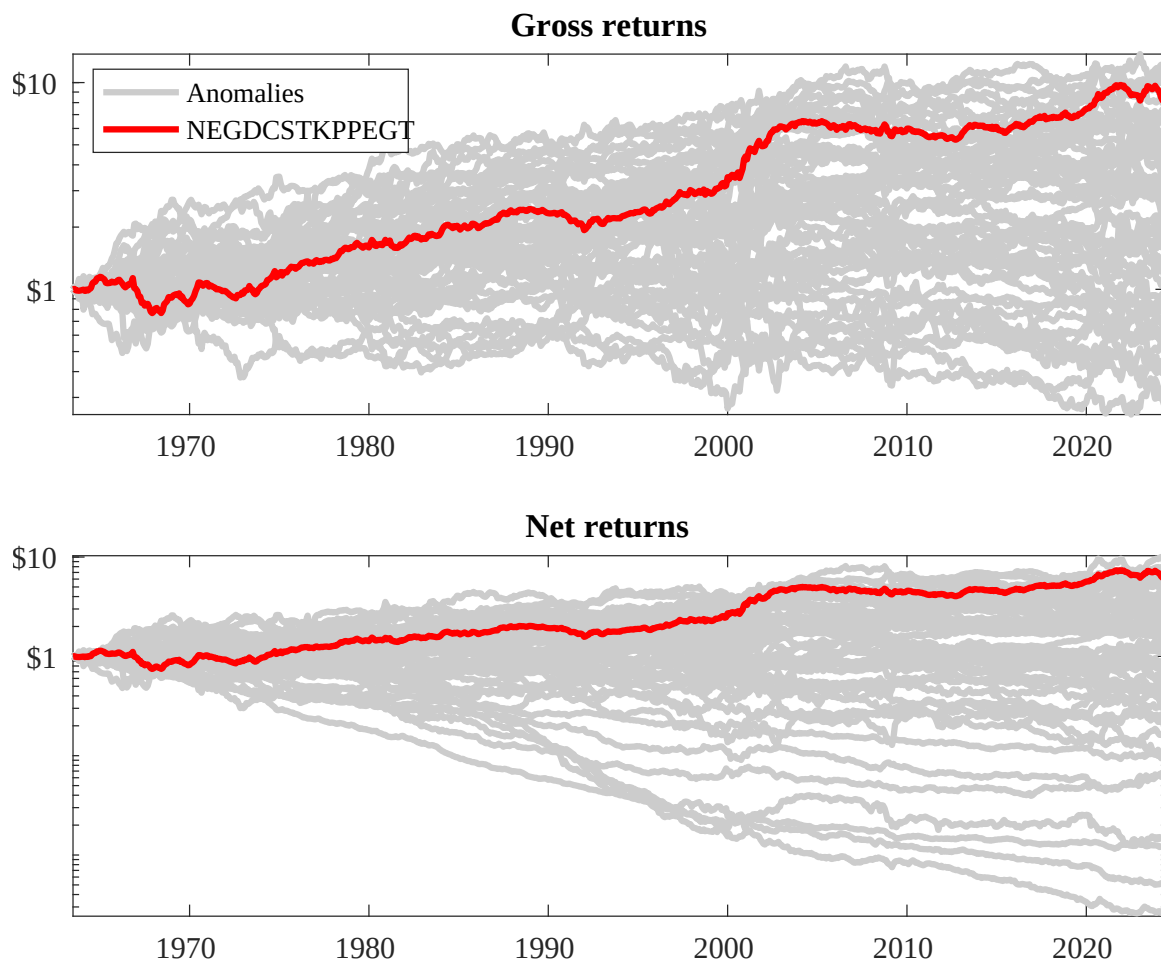
This table presents results for conditional double sorts on size and CSD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CSD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CSD and short stocks with low CSD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                                                    |                 |                |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | CSD Quintiles |                |                |                |                | CSD Strategies                                     |                 |                |                |                |                |                |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |                |
|                                                                       | (1)           | 0.45<br>[1.59] | 0.71<br>[2.69] | 0.85<br>[3.23] | 0.99<br>[4.01] | 1.01<br>[4.30]                                     | 0.58<br>[5.35]  | 0.68<br>[6.34] | 0.57<br>[5.91] | 0.50<br>[5.17] | 0.36<br>[3.95] | 0.31<br>[3.46] |
|                                                                       | (2)           | 0.49<br>[1.98] | 0.76<br>[3.19] | 0.89<br>[3.68] | 0.91<br>[4.05] | 0.95<br>[4.31]                                     | 0.46<br>[4.58]  | 0.55<br>[5.59] | 0.41<br>[4.65] | 0.35<br>[3.90] | 0.32<br>[3.53] | 0.27<br>[3.00] |
|                                                                       | (3)           | 0.65<br>[2.97] | 0.59<br>[2.79] | 0.77<br>[3.49] | 0.79<br>[3.88] | 0.96<br>[4.89]                                     | 0.31<br>[3.45]  | 0.39<br>[4.34] | 0.28<br>[3.38] | 0.24<br>[2.84] | 0.22<br>[2.53] | 0.19<br>[2.17] |
|                                                                       | (4)           | 0.52<br>[2.62] | 0.62<br>[3.06] | 0.81<br>[4.05] | 0.75<br>[3.98] | 0.81<br>[4.45]                                     | 0.28<br>[3.18]  | 0.35<br>[3.96] | 0.23<br>[2.98] | 0.23<br>[2.92] | 0.04<br>[0.52] | 0.06<br>[0.75] |
|                                                                       | (5)           | 0.47<br>[2.83] | 0.54<br>[3.06] | 0.50<br>[2.92] | 0.52<br>[3.18] | 0.72<br>[4.54]                                     | 0.24<br>[2.64]  | 0.27<br>[2.90] | 0.23<br>[2.47] | 0.20<br>[2.15] | 0.21<br>[2.32] | 0.20<br>[2.11] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                                                    |                 |                |                |                |                |                |
| Size quintiles                                                        | CSD Quintiles |                |                |                |                | CSD Quintiles                                      |                 |                |                |                |                |                |
|                                                                       | Average $n$   |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |                |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |                |
|                                                                       | (1)           | 342            | 342            | 342            | 340            | 335                                                | 26              | 29             | 33             | 26             | 24             |                |
|                                                                       | (2)           | 97             | 95             | 96             | 96             | 95                                                 | 50              | 49             | 51             | 49             | 49             |                |
|                                                                       | (3)           | 69             | 69             | 69             | 69             | 69                                                 | 87              | 85             | 87             | 88             | 89             |                |
|                                                                       | (4)           | 60             | 60             | 60             | 60             | 60                                                 | 188             | 189            | 198            | 195            | 199            |                |
| (5)                                                                   | 56            | 56             | 56             | 56             | 55             | 1324                                               | 1387            | 1487           | 1563           | 1734           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

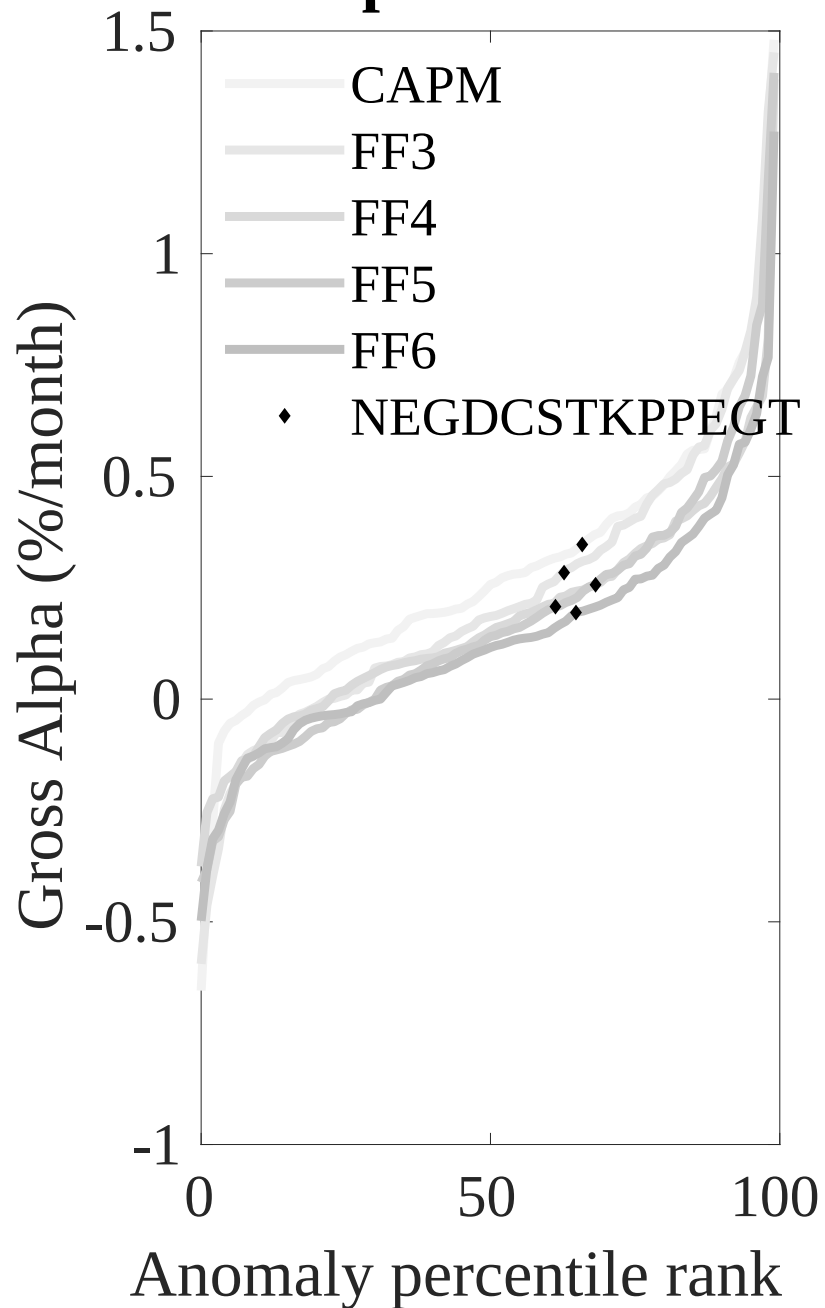
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CSD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

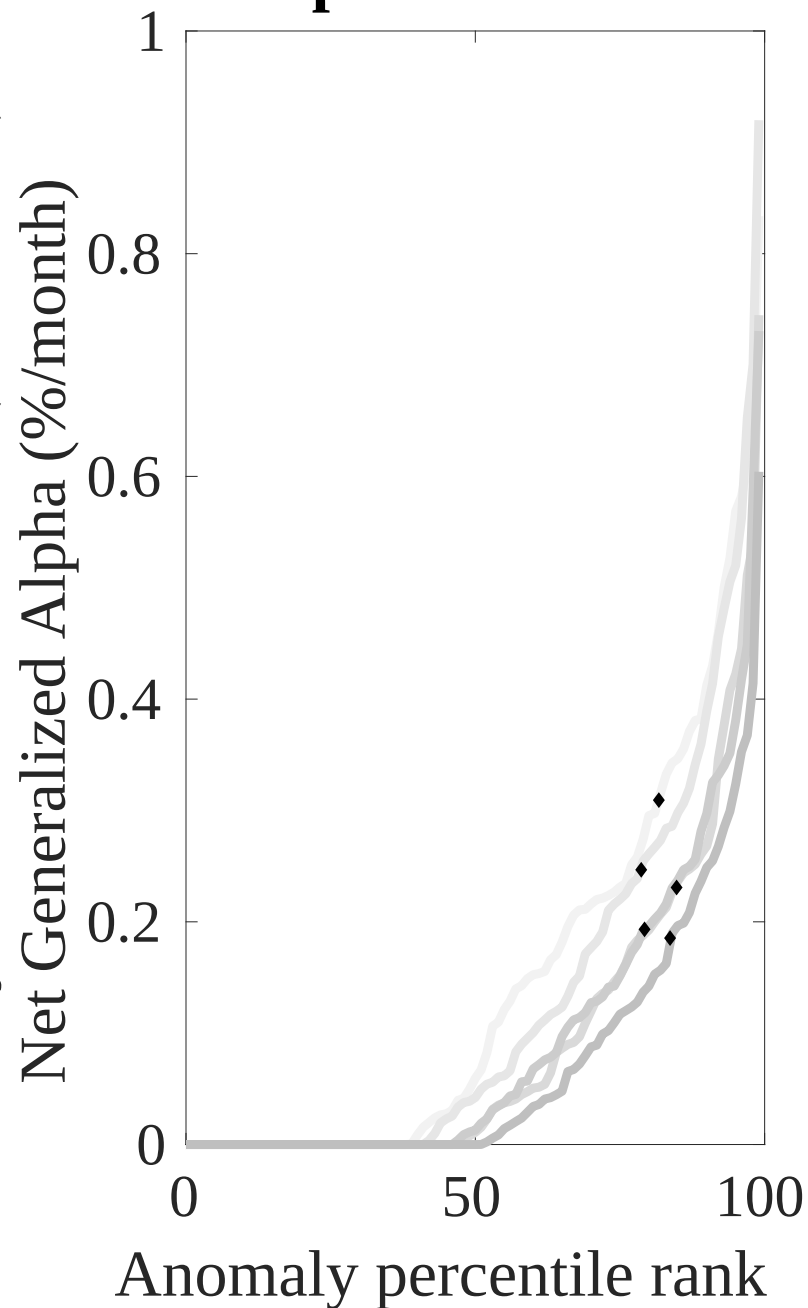
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CSD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

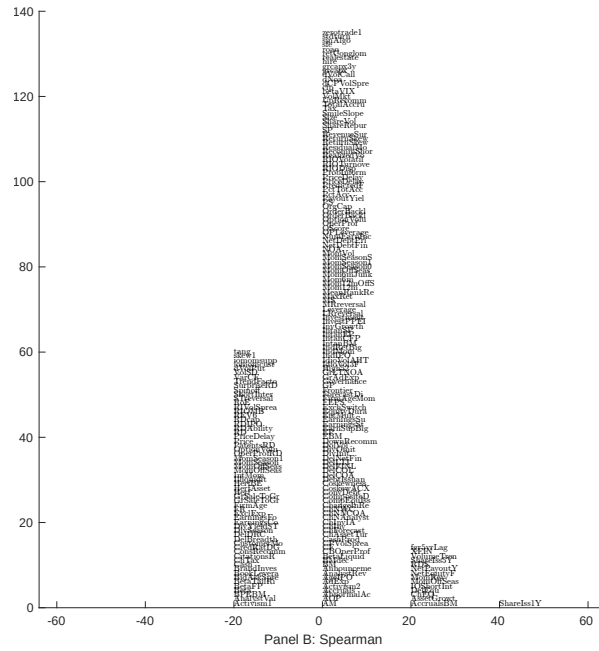
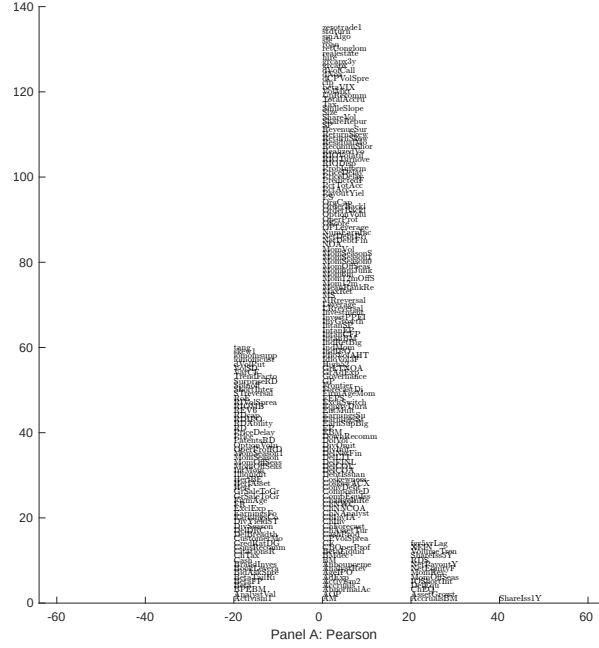
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

## Net Alpha Percentiles

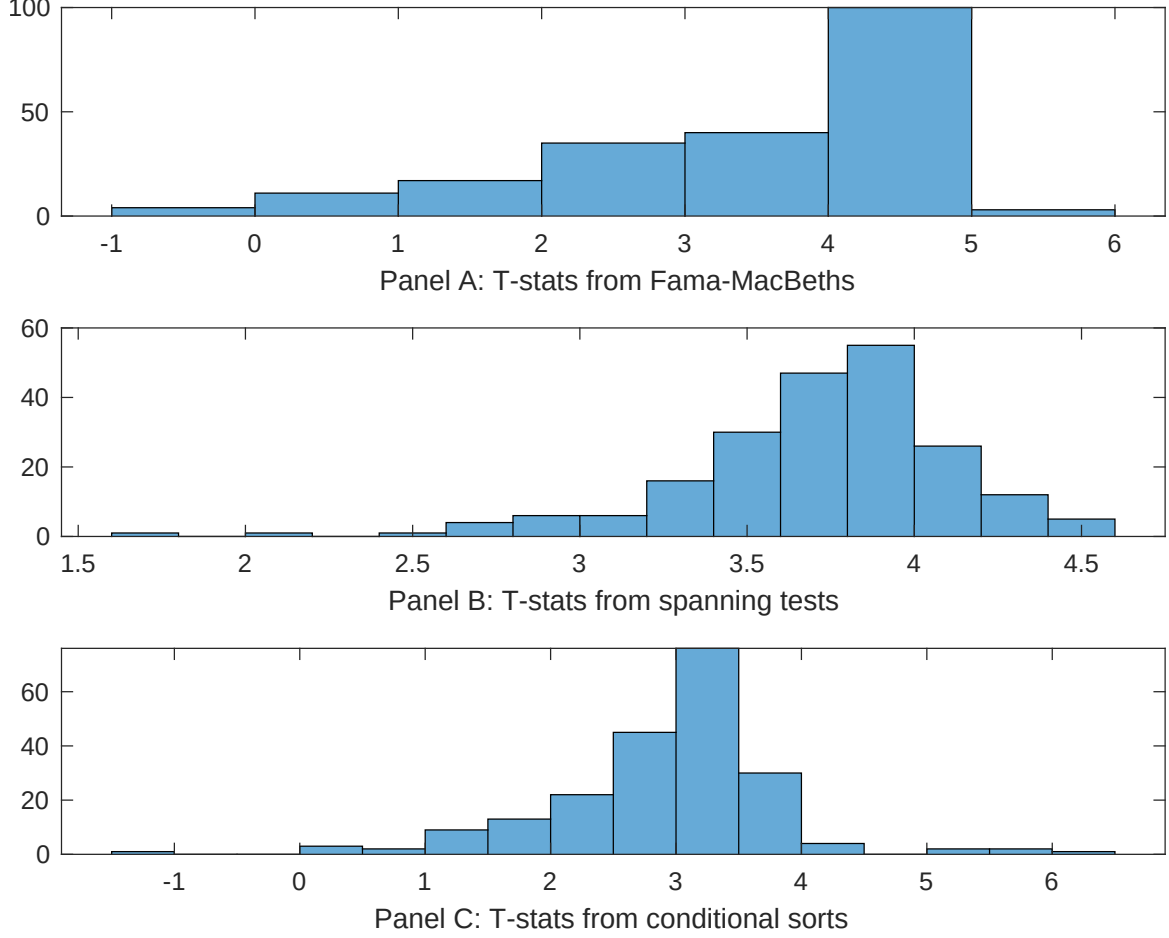




**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CSD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CSD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{CSD}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{CSD}CSD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{CSD,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CSD. Stocks are finally grouped into five CSD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CSD trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

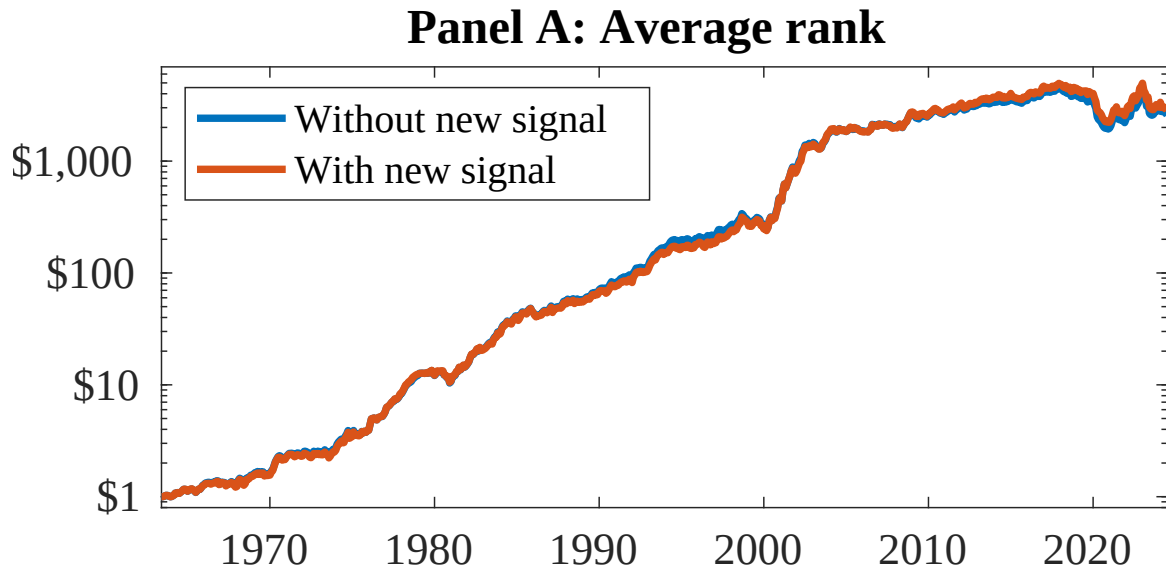
This table presents Fama-MacBeth results of returns on CSD. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{CSD}CSD_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Net Payout Yield, Share issuance (1 year), Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.12<br>[5.42] | 0.13<br>[5.97] | 0.17<br>[6.69] | 0.13<br>[6.36] | 0.13<br>[5.59] | 0.13<br>[6.02] | 0.12<br>[3.87]   |
| CSD             | 0.88<br>[1.71] | 0.17<br>[3.57] | 0.16<br>[3.13] | 0.16<br>[3.39] | 0.19<br>[3.74] | 0.15<br>[3.05] | 0.61<br>[1.19]   |
| Anomaly 1       | 0.22<br>[1.96] |                |                |                |                |                | 1.00<br>[0.94]   |
| Anomaly 2       |                | 0.14<br>[6.18] |                |                |                |                | 0.68<br>[2.48]   |
| Anomaly 3       |                |                | 0.39<br>[3.32] |                |                |                | -0.15<br>[-0.65] |
| Anomaly 4       |                |                |                | 0.28<br>[4.85] |                |                | 0.24<br>[4.23]   |
| Anomaly 5       |                |                |                |                | 0.12<br>[3.26] |                | -0.24<br>[-0.04] |
| Anomaly 6       |                |                |                |                |                | 0.93<br>[7.89] | 0.64<br>[5.39]   |
| # months        | 715            | 715            | 720            | 715            | 720            | 720            | 715              |
| $\bar{R}^2(\%)$ | 1              | 0              | 0              | 1              | 0              | 0              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CSD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{CSD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Net Payout Yield, Share issuance (1 year), Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                   |                   |                  |                  |                  |                  |                   |
|-----------------|-------------------|-------------------|------------------|------------------|------------------|------------------|-------------------|
| Intercept       | 0.23<br>[2.96]    | 0.18<br>[2.36]    | 0.18<br>[2.36]   | 0.13<br>[1.71]   | 0.19<br>[2.49]   | 0.19<br>[2.41]   | 0.15<br>[1.98]    |
| Anomaly 1       | 18.85<br>[6.35]   |                   |                  |                  |                  |                  | 8.64<br>[2.67]    |
| Anomaly 2       |                   | 25.82<br>[6.67]   |                  |                  |                  |                  | 10.24<br>[2.21]   |
| Anomaly 3       |                   |                   | 31.97<br>[7.82]  |                  |                  |                  | 38.06<br>[6.44]   |
| Anomaly 4       |                   |                   |                  | 25.66<br>[6.26]  |                  |                  | 11.70<br>[2.54]   |
| Anomaly 5       |                   |                   |                  |                  | 13.38<br>[3.46]  |                  | -10.85<br>[-2.00] |
| Anomaly 6       |                   |                   |                  |                  |                  | -3.05<br>[-0.65] | -19.69<br>[-3.87] |
| mkt             | 2.87<br>[1.53]    | 0.60<br>[0.33]    | 1.01<br>[0.56]   | 1.12<br>[0.61]   | -0.28<br>[-0.15] | -0.04<br>[-0.02] | 3.45<br>[1.91]    |
| smb             | -0.15<br>[-0.05]  | -2.22<br>[-0.84]  | -4.22<br>[-1.62] | -1.53<br>[-0.58] | -3.32<br>[-1.23] | -2.90<br>[-1.06] | -0.07<br>[-0.03]  |
| hml             | -3.15<br>[-0.84]  | 3.58<br>[1.02]    | 0.35<br>[0.10]   | 6.32<br>[1.79]   | 2.52<br>[0.69]   | 4.44<br>[1.23]   | -1.23<br>[-0.34]  |
| rmw             | -10.09<br>[-2.54] | -13.07<br>[-3.17] | 2.41<br>[0.69]   | -8.28<br>[-2.15] | 2.12<br>[0.58]   | 0.94<br>[0.26]   | -12.88<br>[-3.00] |
| cma             | 20.55<br>[3.59]   | 20.34<br>[3.58]   | 5.47<br>[0.85]   | 23.92<br>[4.32]  | 22.97<br>[3.50]  | 40.13<br>[5.17]  | 15.14<br>[1.95]   |
| umd             | 3.59<br>[1.97]    | 0.91<br>[0.50]    | 1.54<br>[0.86]   | 0.42<br>[0.23]   | 2.44<br>[1.32]   | 2.04<br>[1.10]   | -0.00<br>[-0.00]  |
| # months        | 728               | 728               | 732              | 728              | 732              | 732              | 728               |
| $\bar{R}^2(\%)$ | 20                | 20                | 22               | 19               | 16               | 15               | 27                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CSD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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